

How to Become an AI Engineer in 2026

A free-course roadmap - Python fundamentals to shipping AI agents.

This is a curated, sequenced learning path using only free (or free-to-audit) courses. Roughly 4 months at ~10 hours/week, but you can compress or stretch as needed. For Coursera courses, click "Enroll for Free" and look for the "Audit" option to access videos without paying.

Stage 1 - Python Foundations

Weeks 1–3 · Goal: write Python comfortably, understand functions, OOP, files, and APIs.

CS50P - Introduction to Programming with Python

Provider: Harvard University (edX / free on YouTube)

The gold-standard beginner course. Covers functions, loops, OOP, file I/O, regex, and unit testing. No prior coding experience required. Taught by David Malan.

Link: <https://cs50.harvard.edu/python/>

Scientific Computing with Python

Provider: freeCodeCamp

Project-based and lighter than CS50P. Free certificate included. Good if you want to ship something fast and prefer interactive coding over lectures.

Link: <https://www.freecodecamp.org/learn/scientific-computing-with-python/>

Stage 2 - Math & ML Fundamentals

Weeks 4–7 · Goal: understand how models learn - gradient descent, neural nets, evaluation.

Machine Learning Specialization

Provider: Andrew Ng - Stanford / DeepLearning.AI (Coursera, audit free)

The canonical ML entry point. A 3-course specialization (updated 2022) covering supervised learning, advanced algorithms, unsupervised learning, and recommender systems. Rated 4.9/5 with over 4.8M learners. Click "Audit" on each course to access videos free.

Link: <https://www.coursera.org/specializations/machine-learning-introduction>

Deep Learning Specialization

Provider: Andrew Ng - DeepLearning.AI (Coursera, audit free)

5-course specialization covering CNNs, RNNs, LSTMs, Transformers, plus regularization techniques like Dropout, BatchNorm, and Xavier/He initialization. Uses Python and TensorFlow.

Link: <https://www.coursera.org/specializations/deep-learning>

Stage 3 - LLMs, RAG & the Modern AI Stack

Weeks 8–10 · Goal: build LLM-powered applications - prompting, function calling, RAG, evals.

DeepLearning.AI Short Courses

Provider: DeepLearning.AI (free)

The single best free resource for LLM application engineering. 1–2 hour focused courses on prompt engineering, RAG, function calling, evals, LangChain, vector databases, and more. Pick the ones relevant to your stack.

Link: <https://learn.deeplearning.ai/>

Hugging Face NLP Course

Provider: Hugging Face (free)

The practitioner's path to transformers, tokenization, and fine-tuning. Hands-on with the transformers and datasets libraries.

Link: <https://huggingface.co/learn/nlp-course>

Hugging Face LLM Course

Provider: Hugging Face (free)

Deeper dive into large language models - pretraining, fine-tuning, and deployment.

Link: <https://huggingface.co/learn/llm-course>

Anthropic Cookbook

Provider: Anthropic (free, GitHub)

Reference implementations for tool use, structured outputs, retrieval, sub-agents, and more with Claude.

Link: <https://github.com/anthropics/anthropic-cookbook>

OpenAI Cookbook

Provider: OpenAI (free, GitHub)

Step-by-step guides and example code for everything from embeddings to function calling.

Link: <https://cookbook.openai.com/>

Stage 4 - Building AI Agents

Weeks 11–14 · Goal: build, deploy, and benchmark autonomous agents.

Hugging Face AI Agents Course (Certified)

Provider: Hugging Face (free, with certificate)

The standout free agents course of 2026. Covers the Think → Act → Observe loop, smolagents, LangGraph, and LlamaIndex frameworks. Each chapter is ~3–4 hours over 1 week. You'll build your first agent ("Alfred"), publish it to Hugging Face Spaces, and compete on a public leaderboard.

Link: <https://huggingface.co/learn/agents-course/en/unit0/introduction>

Introduction to LangGraph

Provider: LangChain Academy (free)

Covers cyclic reasoning, state machines, reflection loops, error handling, and workflow orchestration. The essential resource for stateful agent patterns.

Link: <https://academy.langchain.com/courses/intro-to-langgraph>

AI Agent Developer Specialization

Provider: Vanderbilt University - Dr. Jules White (Coursera, audit free)

6-course specialization (~64 hours) covering agent architecture, tool use, memory, and multi-agent systems. Recommended starting order: AI Agents & Agentic AI with Python first, then Prompt Engineering for ChatGPT.

Link: <https://www.coursera.org/specializations/ai-agents>

Stage 5 - MCP & Production

Week 15+ · Goal: connect agents to real-world tools and ship production-grade systems.

Model Context Protocol (MCP) Documentation

Provider: Anthropic (free)

MCP is the connective tissue for 2026 agents - a standard protocol for connecting LLMs to tools and data sources. Start with the official spec and quickstart.

Link: <https://modelcontextprotocol.io/>

MCP: Build Rich-Context AI Apps with Anthropic

Provider: DeepLearning.AI x Anthropic (free)

Short course (~1.5 hours) covering MCP fundamentals - building servers, clients, and integrating with Claude.

Link: <https://www.deeplearning.ai/short-courses/mcp-build-rich-context-ai-apps-with-anthropic/>

Claude Cookbook

Provider: Anthropic (free, GitHub)

Production patterns: building agents with Claude, computer use, sub-agents, and MCP integration examples.

Link: <https://github.com/anthropics/claude-cookbooks>

How to Use This Roadmap

- Sequence matters - Stages 1 and 2 are non-negotiable foundations. Skipping them means hitting a wall later.
- For Coursera specializations, the "Audit" link is hidden - click "Enroll for Free" first, then look for it in the modal. Videos are free; graded assignments and certificates are paid.
- The Hugging Face Agents course is the centerpiece - it's free, certified, modern, and includes a public leaderboard challenge.
- Don't just watch - build alongside. Learners who ship a small working agent within 48 hours of starting are far more likely to keep going.
- MCP is the 2026 differentiator. Most developers haven't touched it yet - getting fluent here is high-leverage.